

McIntyre’s Fitness App

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Project Description

McIntyre’s Fitness App is a Windows Forms desktop application built using C# and MySQL. It allows users to register accounts, log workouts, view their training history, edit and delete entries, and track progress statistics such as total workouts, total weight lifted, heaviest lift, and favorite exercise.

The application uses a three-layer architecture separating the user interface, business logic, and MySQL database. All data is stored persistently so users can track progress across sessions.

Project Justification

Many people struggle to maintain consistent workout records and measure their progress over time. Existing fitness apps are often mobile only or require subscriptions. McIntyre’s Fitness App provides a free, local desktop solution that gives users full control of their data without relying on cloud services or internet connectivity.

The app demonstrates core software engineering principles including CRUD operations, user authentication, session management, and data security through SHA256 password hashing.

Functional Requirements

- FR-001:** Users can register a new account
- FR-002:** Users can log in securely with hashed passwords
- FR-003:** Users can log workouts with exercise, sets, reps, and weight
- FR-004:** Users can view their full workout history
- FR-005:** Users can edit existing workout entries
- FR-006:** Users can delete workouts with confirmation dialog
- FR-007:** Users can view progress statistics
- FR-008:** Users can log out and clear their session

Product Specifications

- Language:** C# (.NET 9)
- UI Framework:** Windows Forms
- Database:** MySQL (local)
- IDE:** Visual Studio 2022
- Security:** SHA256 password hashing, parameterized SQL queries
- Version Control:** GitHub (github.com/JohnPMcIntyre/Capstone)

Design Process

This project followed an Agile Scrum methodology across two classes (CST-451 and CST-452). Development was organized into iterative milestones with user stories, sprint planning, and continuous testing.

Phase 1 - Planning: Project proposal, requirements specification, and architecture design including ER diagrams, sequence diagrams, and database schema.

Phase 2 - Development: Iterative coding using C# Windows Forms with MySQL integration. Features were built and tested incrementally across Milestones 4 and 5.

Phase 3 - Testing: Unit and system testing performed on all 10 modules with both positive and negative test cases documented in the official GCU test case template.

Relevant Data Summary

User Stories Completed: 11 of 12 (92%)
Total Development Hours: 33 estimated / 35 actual
Test Cases Executed: 40+ across 10 modules
Overall Pass Rate: Approximately 95% (registration error handling identified and resolved)
Features Implemented: Registration, Login, Workout CRUD, Progress Tracking, Logout, Password Hashing, Delete Confirmation

All source code is version controlled and available at github.com/JohnPMcIntyre/Capstone



WorkoutForm

MCINTYRE FITNESS

Exercise Name

Number of Sets

0

Number of Rep

0

Number of Weight

0

Save Workout

Back

